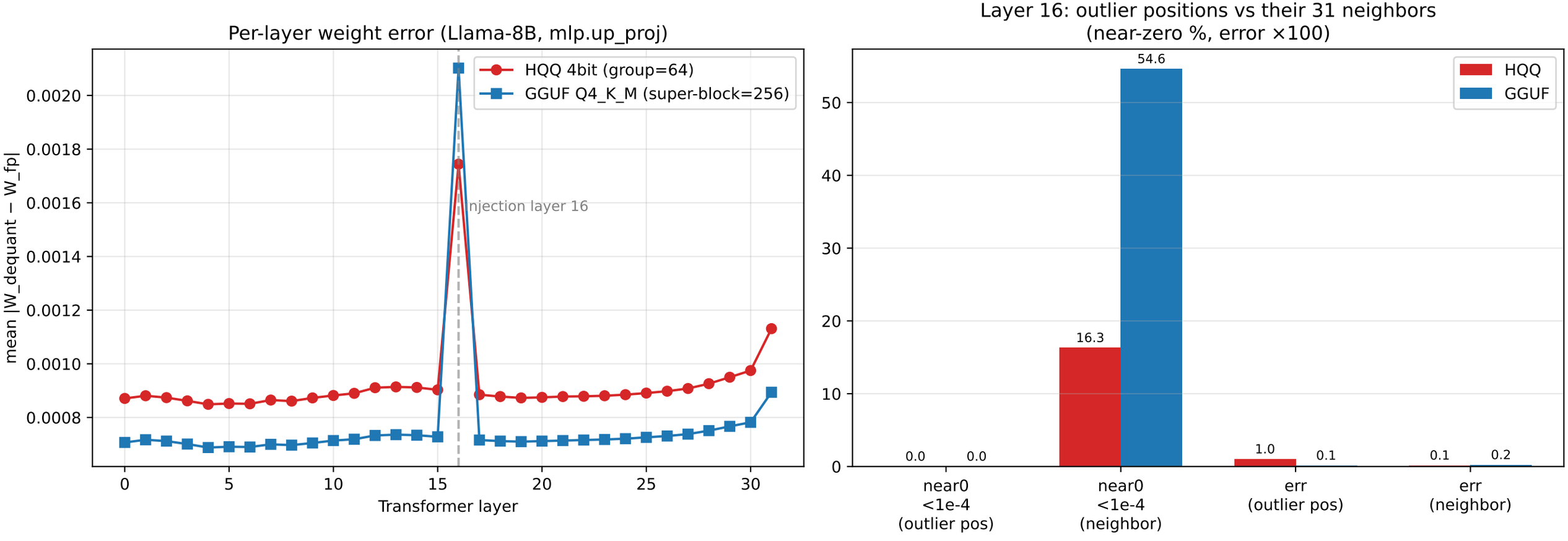
Per-layer weight error (Llama-8B, mlp.up_proj)
HQQ 4bit (group=64)
GGUF Q4_K_M (super-block=256)
injection layer 16
mean |W_dequant − W_fp|
Transformer layer
0.0008
0.0010
0.0012
0.0014
0.0016
0.0018
0.0020
0
5
10
15
20
25
30
Layer 16: outlier positions vs their 31 neighbors
(near-zero %, error ×100)
HQQ
GGUF
0.0
0.0
16.3
54.6
1.0
0.1
0.1
0.2
0
10
20
30
40
50
near0
<1e-4
(outlier pos)
near0
<1e-4
(neighbor)
err
(outlier pos)
err
(neighbor)